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Course code:DSA0602

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Scenario 1: Student Academic Performance Dashboard

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("scenario1_student_academic_performance.csv")

plt.figure(figsize=(8,5))

sns.scatterplot(data=df,
                x="Attendance (%)",
                y="Internal Marks",
                hue="Department",
                s=120)

plt.title("Student Academic Performance Dashboard")

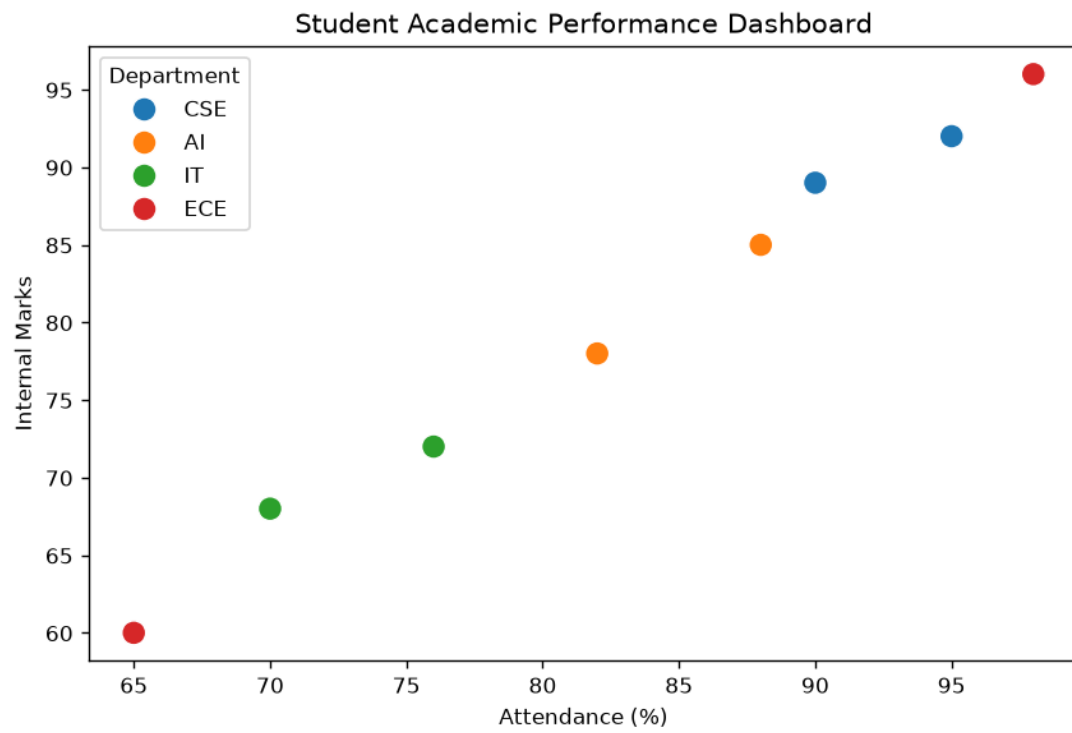
plt.xlabel("Attendance (%)")

plt.ylabel("Internal Marks")

plt.legend(title="Department")

plt.show()
```

Output:



Scenario 2: Electric Vehicle Market Analysis

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

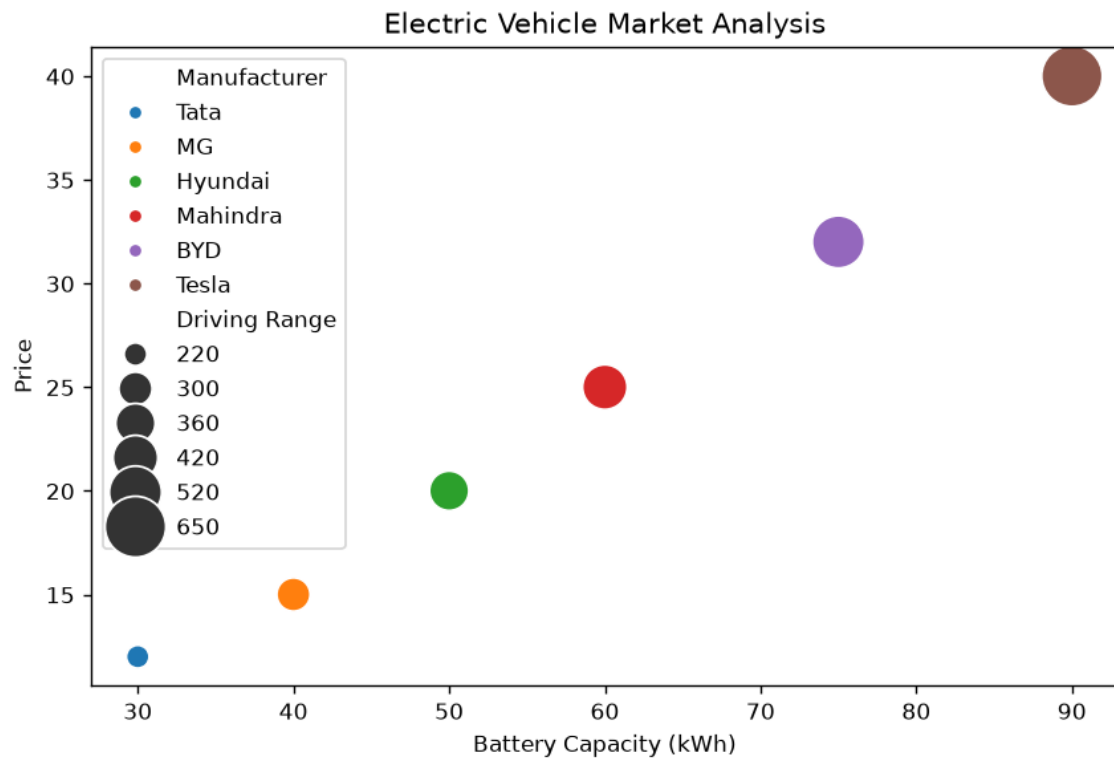
df = pd.read_csv("scenario2_ev_market(1).csv")

plt.figure(figsize=(8,5))

sns.scatterplot(data=df,
                x="Battery Capacity (kWh)",
                y="Price",
                hue="Manufacturer",
                size="Driving Range",
                sizes=(100,700))

plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price")
plt.show()
```

Output:



Scenario 3: Hospital Patient Admission Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("scenario3_hospital_admissions.csv")

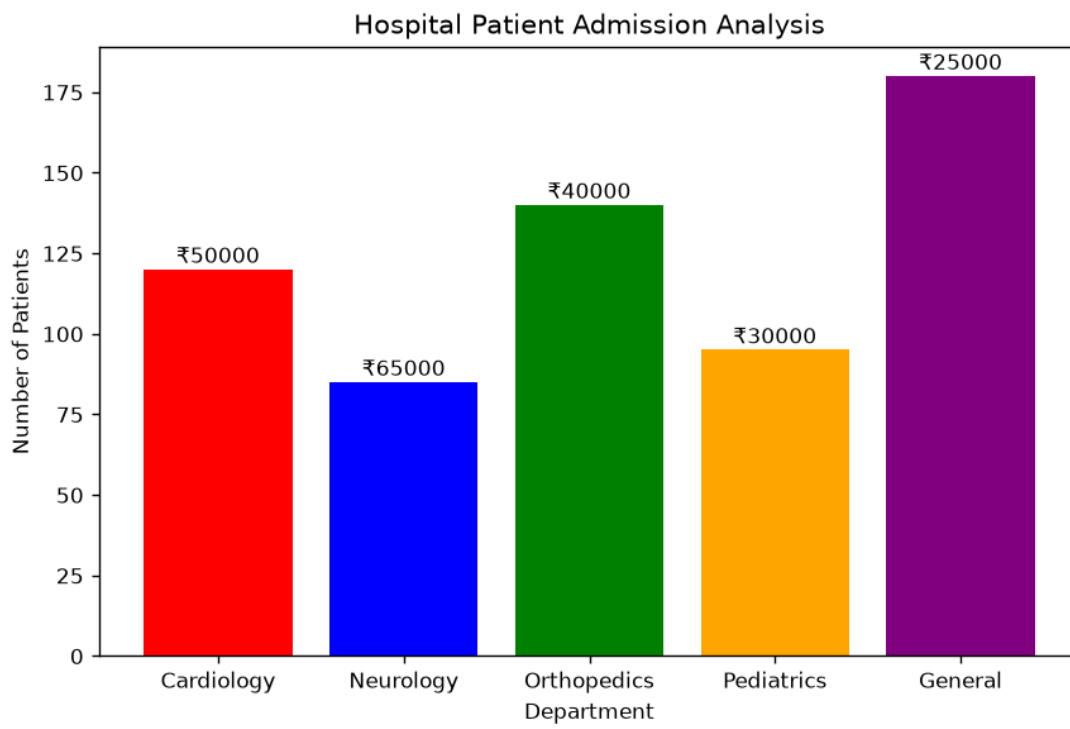
plt.figure(figsize=(8,5))

bars = plt.bar(df["Department"],
               df["Number of Patients"],
               color=["red", "blue", "green", "orange", "purple"])

for bar, cost in zip(bars, df["Average Treatment Cost"]):
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height()+2,
             f'₹{cost}',
             ha="center")

plt.title("Hospital Patient Admission Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")
plt.show()
```

Output:



Scenario 4: Weather Monitoring System

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("scenario4_weather(1).csv")

plt.figure(figsize=(9,5))

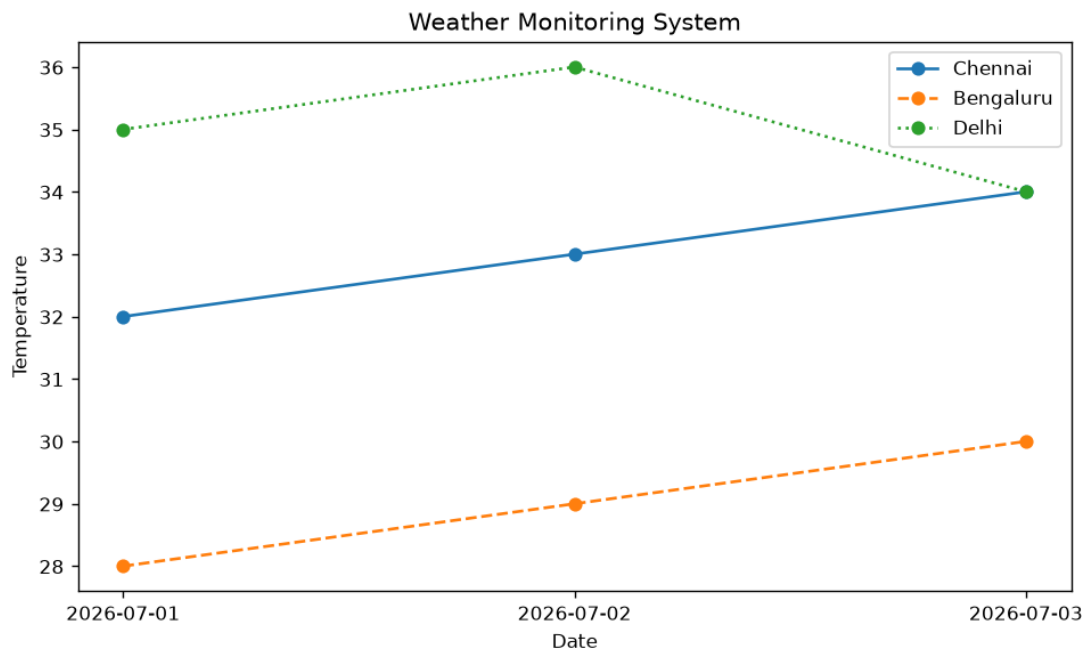
styles = ["-", "--", ":"]

for i,city in enumerate(df["City"].unique()):
    temp = df[df["City"]==city]
    plt.plot(temp["Date"],
             temp["Temperature"],
             linestyle=styles[i],
             marker="o",
             label=city)

plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature")
plt.legend()
```

```
plt.show()
```

Output:



Scenario 5: Supermarket Sales Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

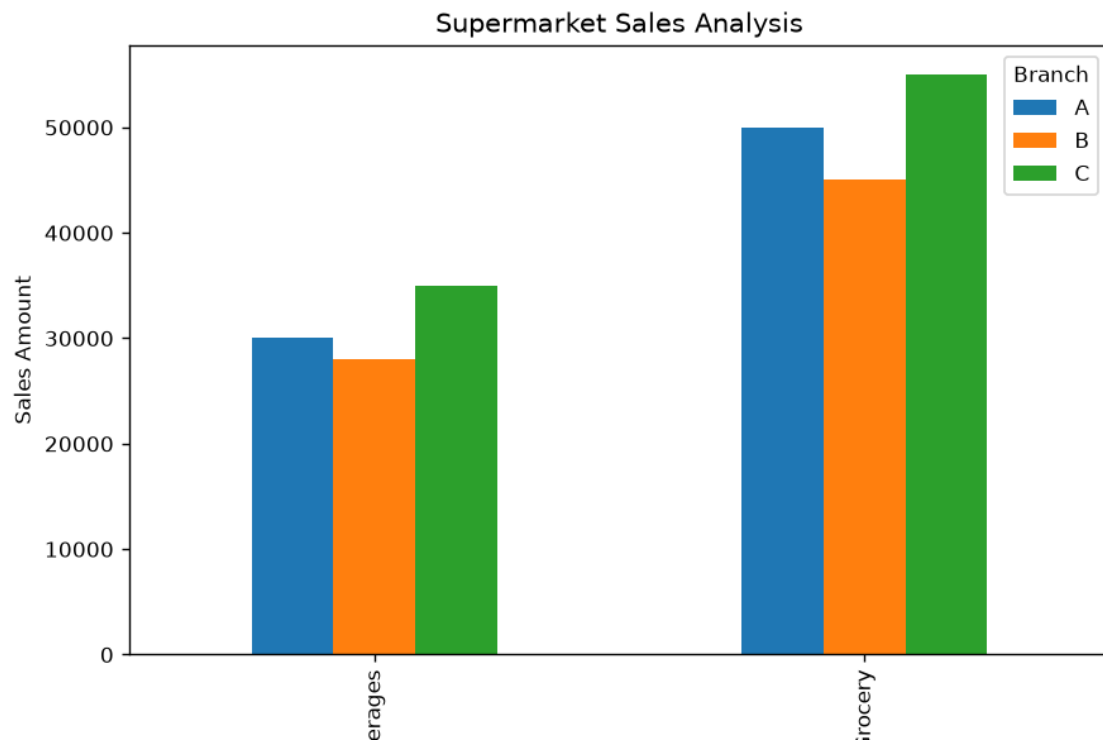
df = pd.read_csv("scenario5_supermarket_sales(1).csv")

pivot = df.pivot(index="Product Category",
                  columns="Branch",
                  values="Sales Amount")

pivot.plot(kind="bar",figsize=(8,5))

plt.title("Supermarket Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales Amount")
plt.show()
```

Output:



Scenario 6: Employee Salary Distribution

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

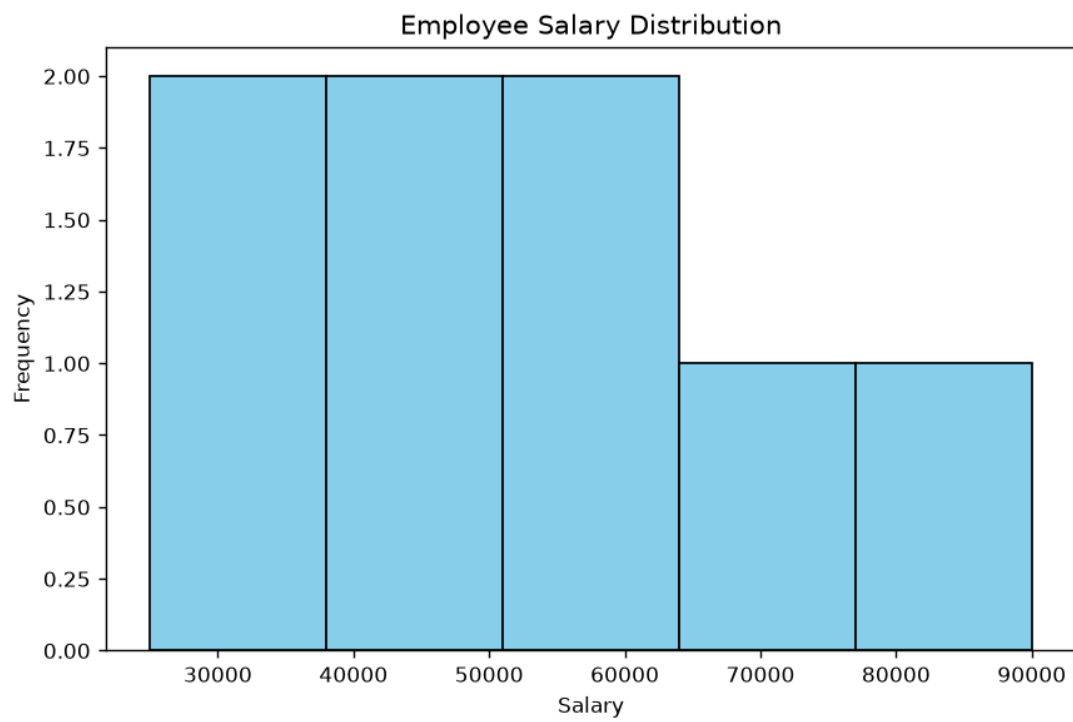
df = pd.read_csv("scenario6_salary(1).csv")

plt.figure(figsize=(8,5))

plt.hist(df["Salary"],
         bins=5,
         color="skyblue",
         edgecolor="black")

plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()
```

Output:



Scenario 7: Smartphone Market Comparison

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("scenario7_smartphones(1).csv")

plt.figure(figsize=(8,6))

scatter = plt.scatter(df["RAM"],
                      df["Price"],
                      s=df["Storage"],
                      c=df["Processor Speed"],
                      cmap="viridis")

for i,brand in enumerate(df["Brand"]):
    plt.text(df["RAM"][i],
             df["Price"][i],
             brand)

plt.colorbar(scatter,label="Processor Speed")

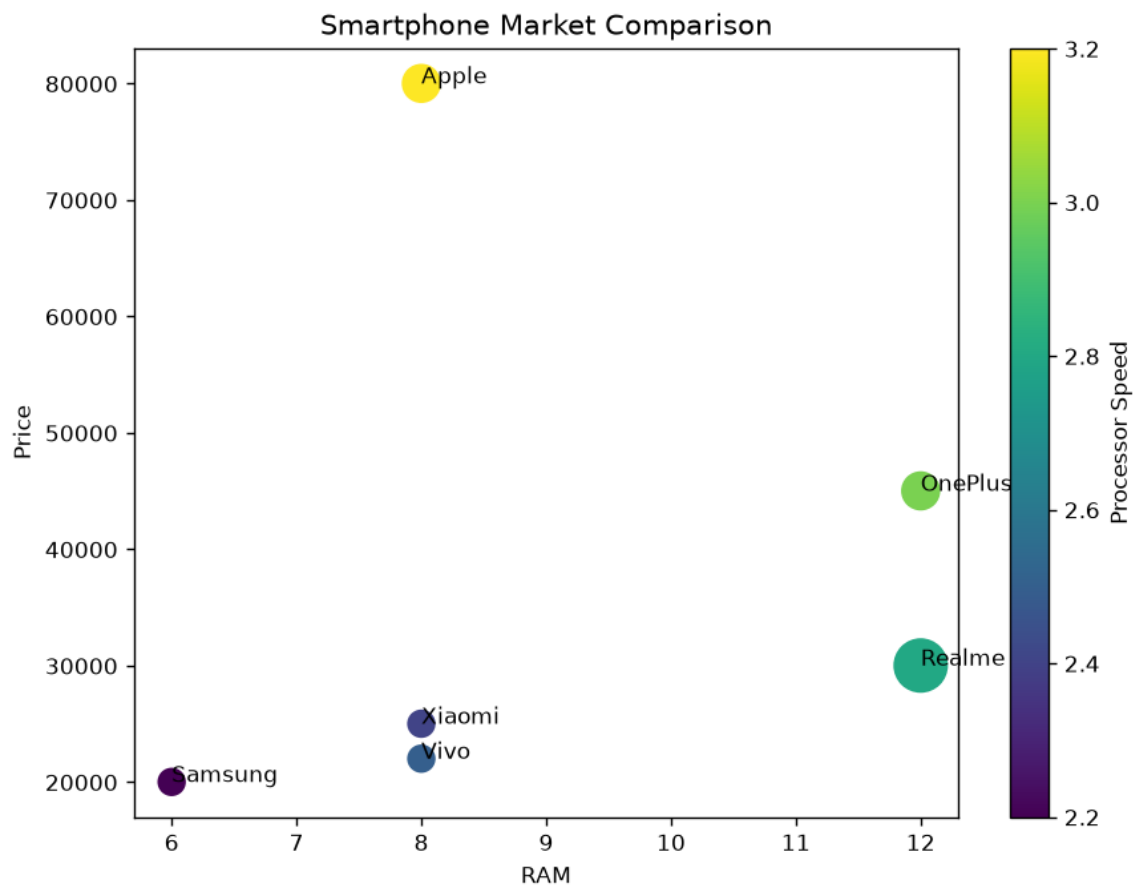
plt.title("Smartphone Market Comparison")
```

```
plt.xlabel("RAM")
```

```
plt.ylabel("Price")
```

```
plt.show()
```

Output:



Scenario 8: Online Movie Rating Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("scenario8_movie_ratings.csv")

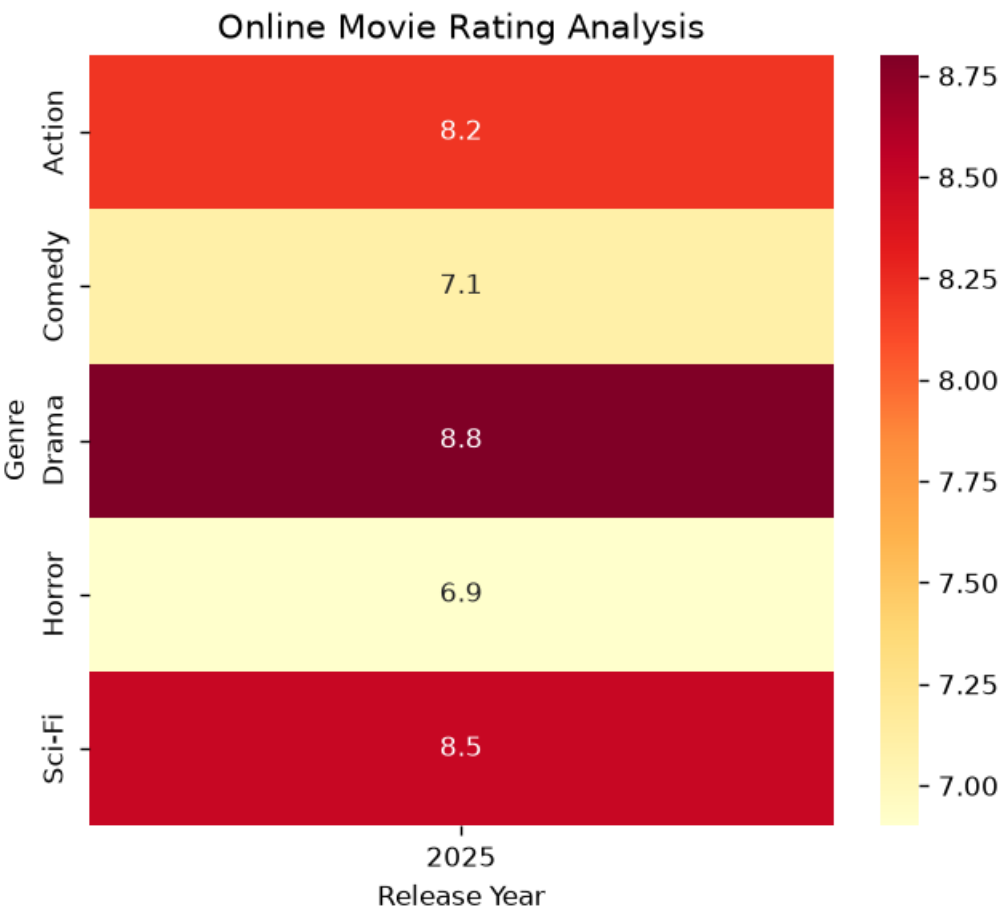
pivot = df.pivot(index="Genre",
                  columns="Release Year",
                  values="Average Rating")

plt.figure(figsize=(6,5))

sns.heatmap(pivot,
            annot=True,
            cmap="YlOrRd")

plt.title("Online Movie Rating Analysis")
plt.show()
```

Output:



Scenario 9: COVID-19 State-wise Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

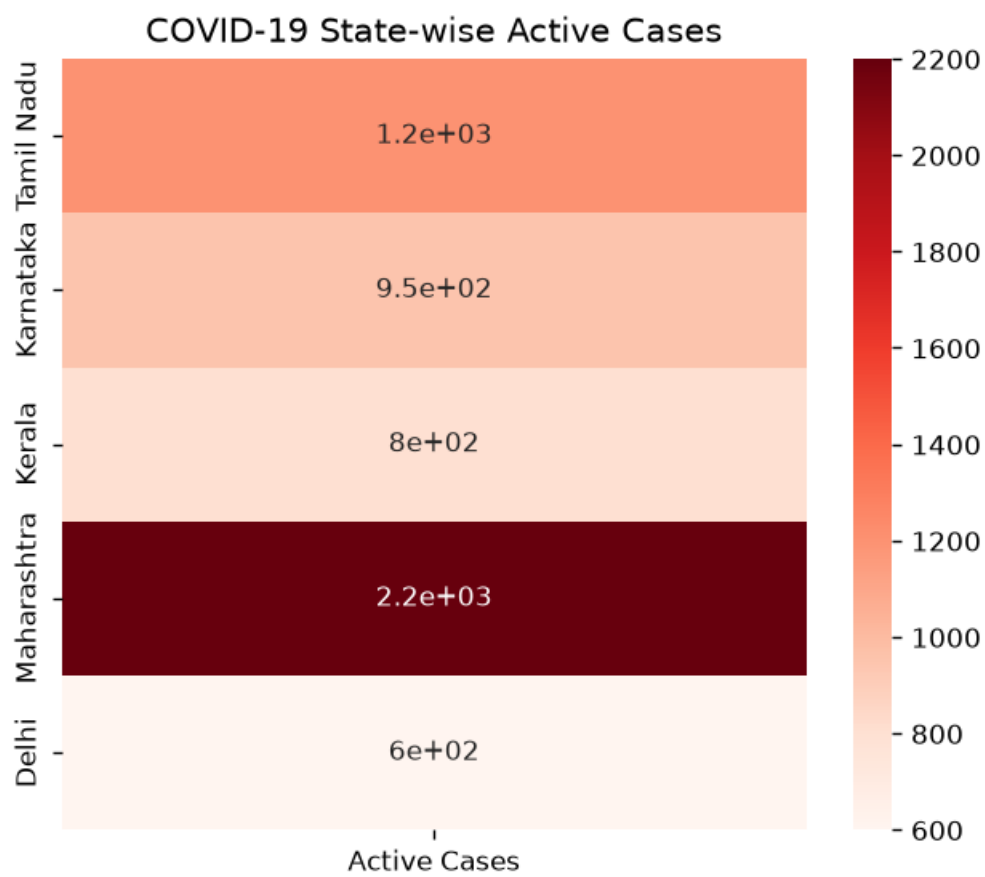
df = pd.read_csv("scenario9_covid_states.csv")

plt.figure(figsize=(6,5))

sns.heatmap(df[["Active Cases"]],
            annot=True,
            yticklabels=df["State"],
            cmap="Reds")

plt.title("COVID-19 State-wise Active Cases")
plt.show()
```

Output:



Scenario 10: Iris Flower Classification

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("scenario10_iris_sample(1).csv")

plt.figure(figsize=(8,6))

sns.scatterplot(data=df,
                x="Sepal Length",
                y="Petal Length",
                hue="Species",
                style="Species",
                size="Sepal Width",
                sizes=(50,300))

plt.title("Iris Flower Classification")

plt.xlabel("Sepal Length")

plt.ylabel("Petal Length")

plt.show()
```

Output:

